



Improving Users' Experiences with Security and Privacy Tasks on Mobile Devices

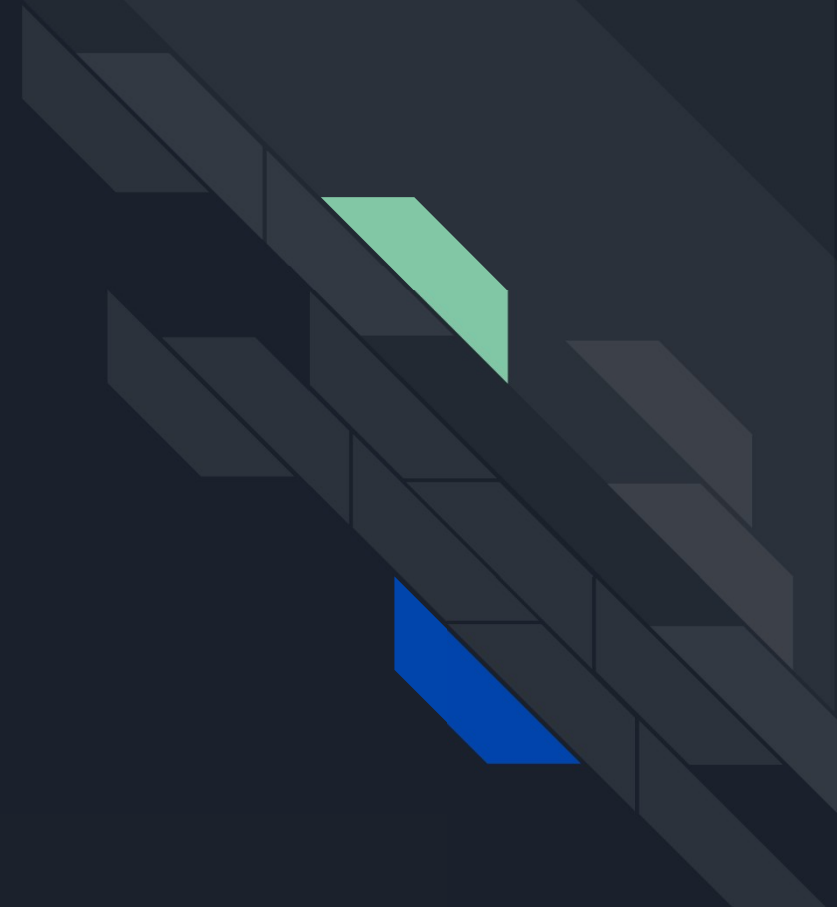
 Brian Mellinger

Slide 1

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<https://docs.google.com/document/d/1yQOfHeTTz1saXFCC0AfgLAlbCXuCtQbX4THYhygupqM/edit?usp=sharing>
Cori Faklaris, 2/17/2026

About me



Smishing is a Usability and Security Challenge

1. The study
 - a. Smishing overview
 - b. Solutions



Courtesy of Gemini - "Create a graphic of google scholar, then zotero, then drive, the sheets



Smishing is not Random, its Coordinated and Structured

1. Coordinated ecosystem Threat

- a. URLs and Shorteners (sanjari et al., 2025)
- b. Brand impersonation (sanjari et al., 2025)
- c. Cross channel coordination (social media, calls, emails, text) (tabassum et al.)

1. Japan study (saeki et al., 2022)

- a. Circadian rhythms
- b. Updates twice a day
- c. Common domain registers used (godaddy)



Human Psychology Drives Smishing Success

1. Psychological manipulation

- a. Authority impersonation (Tabassum et al.)
 - i. IRS, DHS, housing, immigration
- b. Fear, urgency, greed used (zimba et al., 2025) (Rahman et al., 2025)

1. Cultural and contextual Vulnerabilities

- a. Education migrants (Tabassum et al.)
- b. Race (black, mexican-american) (Rahman et al., 2023)
- c. Gender (men) (Yeboah -Boateng & amanor, 2014)
- d. Users without cybersecurity training (Tabassum et al., 2024)
- e. Zambia study (63% unaware) (Zimba et al., 2025)



Detection Alone Cannot Keep up with Evolving Smishing Campaigns

1. Machine learning approaches

- a. S-Detector - Naive Bayes classifier (Joo et al., 2017)
- b. SmiDCA - Statistical correlator and noun classifier (Sonowal & Kuppusamy, 2018)

1. Limitations

- a. URL changes rapidly (Joo et al., 2017)
- b. Mobile carrier ineffectiveness (25-35% blocked) (Commercial Anti-Smishing Tools, 2023)
- c. Lack of reporting of smishing messages (Tabassum et al., 2024) (Faklaris, 2024)

Can be described as a cat and mouse issue.

Educational problem.



Effective Defense Requires Education

1. Focus on Education (Yeboah-Boateng & Amanor, 2014) (Njunga et al., 2022)
 - a. Issues with ...
 - i. TCPA
 - ii. CAN-SPAM act
 - iii. TRACED act

1. Tradition methods are not working (Sonowal & Kuppusamy, 2018)
 - a. Blacklist
 - b. Whitelist
 - c. Carrier level blocking

1. Machine learning (Sanjari et al, 2025) (Seo et al., 2024)
 - a. Needs full-time professional paid help with governmental support



Questions?